

Plantir

An IoT plant management solution

Team Members

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Advisor

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Introduction

- Plantir is an IoT solution for plant care
 - Monitor factors that effect plant health with microcontrollers and sensors
 - Help make educated decisions regarding plant health
 - Simple, cheap as possible

Problem

There should be a cheap, easy way to optimally care for your plants, especially when away.

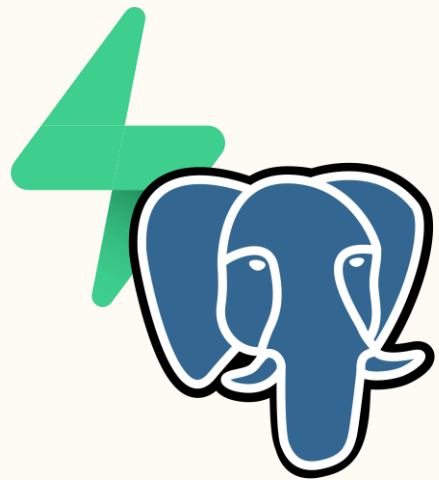
Goal

Show technical depth and simulate a real-world product

User Stories

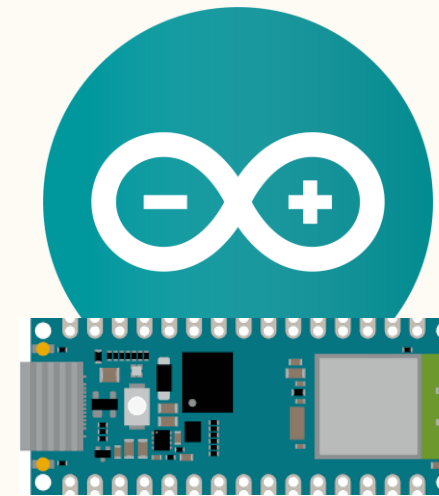
- As a plant owner, I want real time metrics of my plants and their optimal ranges so that I can make better decisions about plant health.
- As a traveler, I want the app to automatically water a plant or water when I trigger it so that my plants can stay hydrated when I'm away.
- As an enthusiast, I want a simple microcontroller setup linked to a certain plant so that I can easily and seamlessly expand my garden monitoring system.

Stack



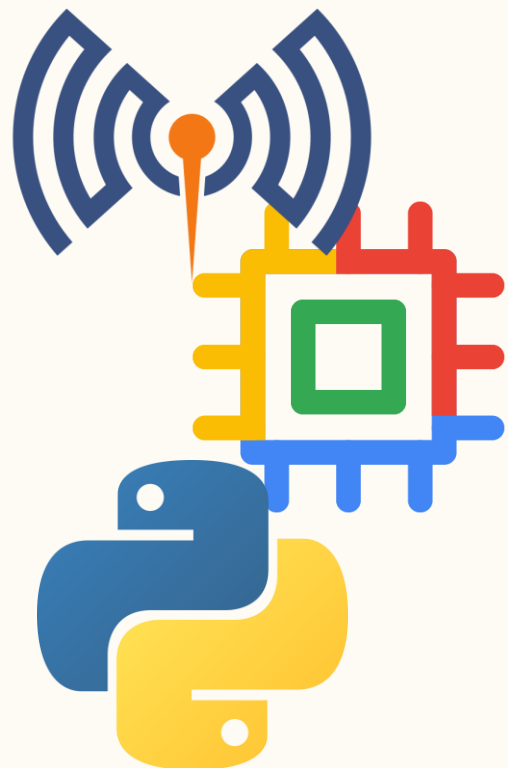
Supabase (PostgreSQL)
database

- RLS
- User Authentication
- User Data



Arduino ESP32 Nano

- Sensor data publishing
- Action on command



Google Compute Engine

(e2-medium)

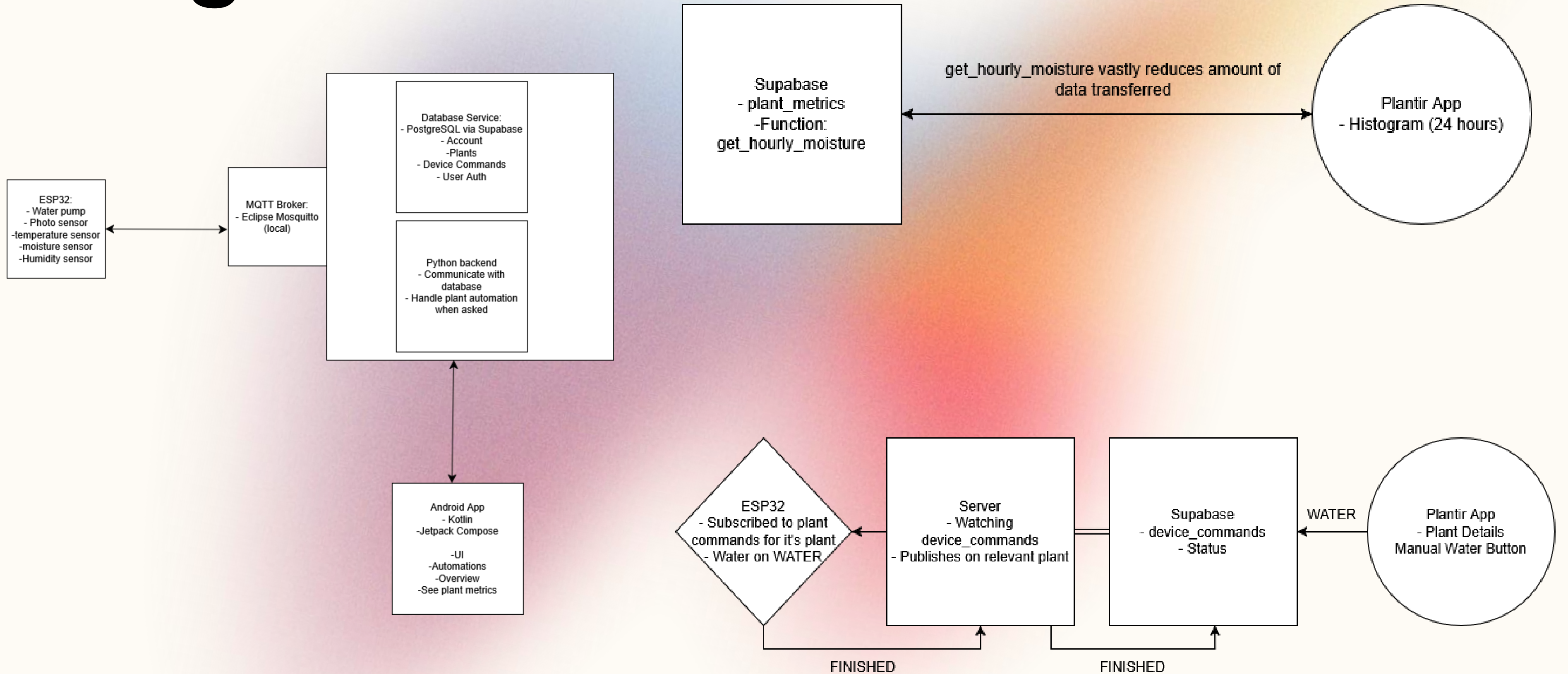
- Mosquitto (MQTT Broker)
- Python
 - Automation
 - Command, message passing



Native Android App

- Kotlin
- Jetpack Compose
 - Material Design System
- User Interface

Diagrams



```
[COMMAND DISPATCHED] -> WATER to device-48CA432F5CE8 on users/3d2ccd20-09e1-449a-bf6e-f415aeab984f/plants/device-48CA432F5CE8/commands
```

Scope and Limitations

Time Constraints

- Both have been working full time past 2 semesters
- 3D printing takes time

Cost Constraints

- Microcontroller, sensors, and server are all paid for
- User needs a 3D printer / order a print

Scope Creep

- Preventing project scope creep by keeping app standalone (i.e. web firmware flashing)



Setup

Unified Enclosure

- Arduino Nano ESP32
- Soil Moisture Capacitive Sensor
- Air / Environment Humidity Sensor
- Temperature Sensor
- Photo Sensor
- Water Reservoir and Pump

Features Planned



Finalized Unified Enclosure

Sealed reservoir,
New light sensor position,
New soil sensor position



Microcontroller Provisioning

SoftAP or Bluetooth,
Pre-flashed or web flash
firmware



Plant API Implementation

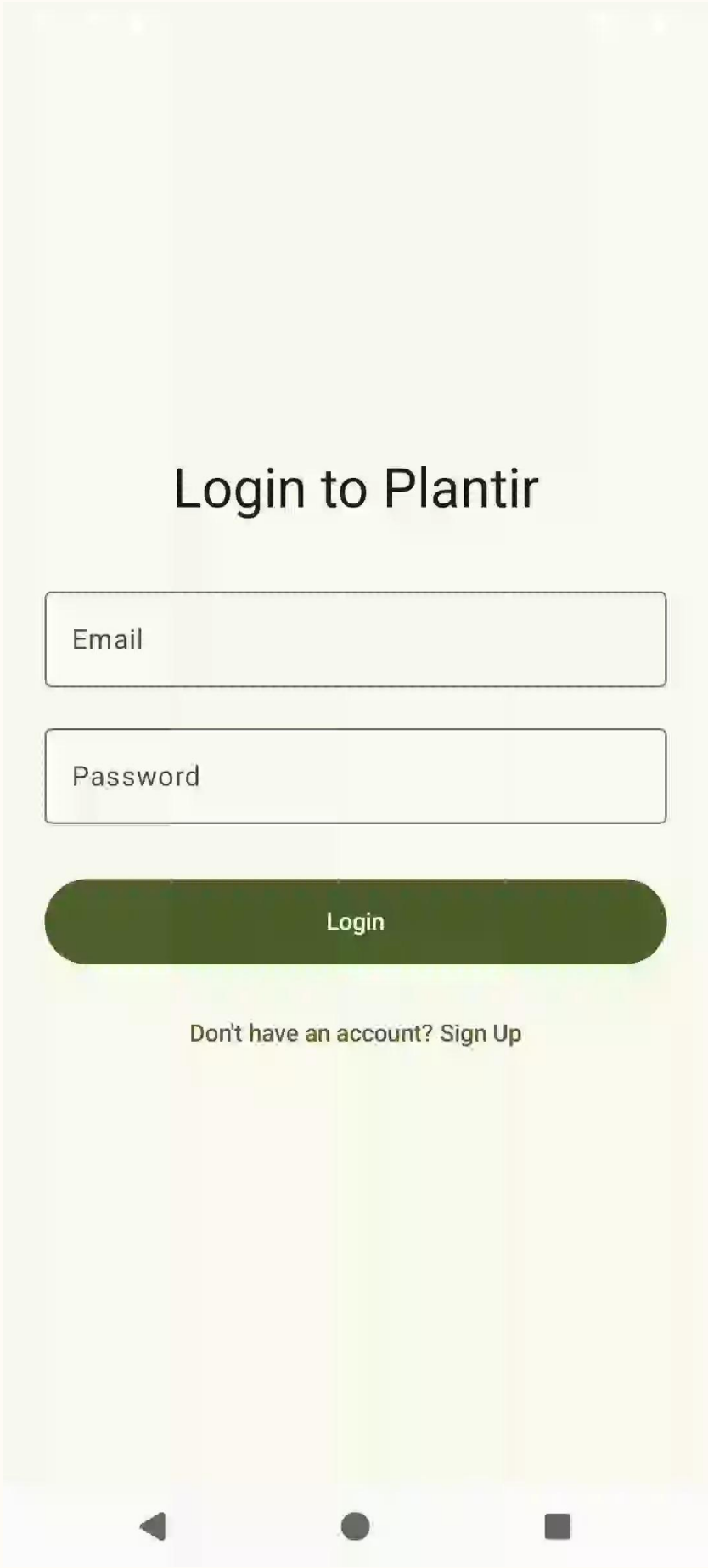
Link Plant API / local
database to plants



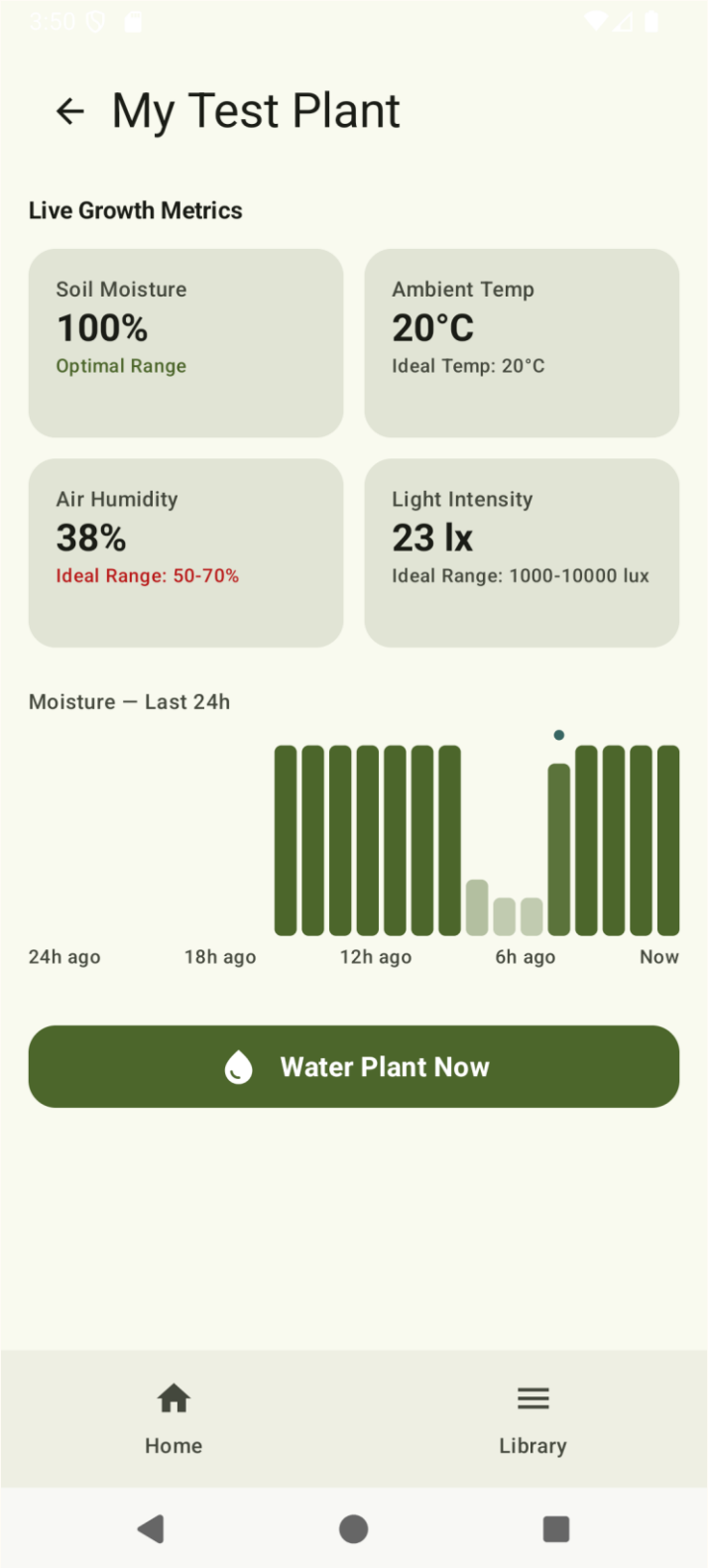
Care plans and automation

Suggest care plans for user
to follow or app to
automate

Recorded Demo



Varied Histogram



Q&A

Thank you!